DAY 2:

Linear Independence, Basis Vectors

Every vector contributes a new direction/information.

A group of vectors is **linearly independent** if every vector gives us a new direction and none of the vectors can be made using the others.

Mathematically, vectors $v_1, v_2, \ldots, v_n$ are linearly independent if the equation

$$c_1v_1 + c_2v_2 + \cdots + c_nv_n = 0$$

has **only the trivial solution**: $c_1 = c_2 = \ldots\ldots c_n = 0$

Example1: Consider: $u = (1, 0)$ and $v = (0, 1)$

(u points right, v points up. These are 2 different directions. You cannot create (0, 1) using only (1, 0). So they are linearly independent)

OR Check: $au + bv = (0, 0)$

au + bv = $a(1, 0) + b(0, 1) = (0, 0)$ So, $(a, b) = (0, 0)$

The only solution is: $a = 0,\ b = 0$

Hence: $\boxed{u, v \text{ are linearly independent}}$ They point in **different directions**.

Example 2: Consider: $u = (1, 2)$ and $v = (2, 4)$

Notice: $v = 2u$, So one vector can be obtained from the other.

For example: $2u - v = 0$ i.e. 2(u) – 1(v) = 0, comparing with au + bv =0

Here, we have a **non-zero solution**: $a = 2,\ b = -1$

Therefore: **u, v are linearly dependent**.

If one vector is simply a multiple of another: $\boxed{v = cu}$ then the vectors are **linearly dependent**.

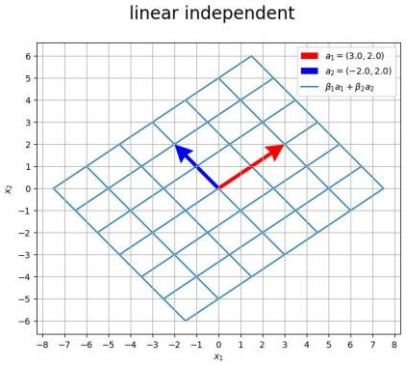
For two vectors in 2D:

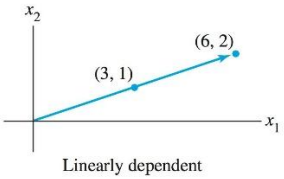
Linearly independent

The vectors point in different directions. They form a **2D plane** when combined.

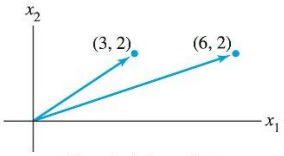
Linearly dependent

The vectors lie on the **same line**.

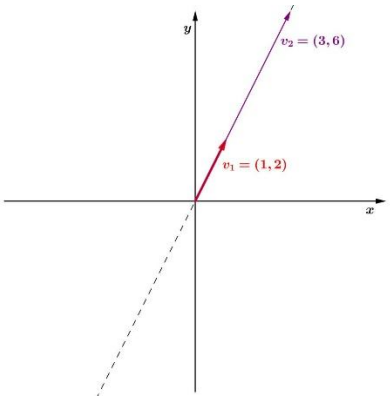


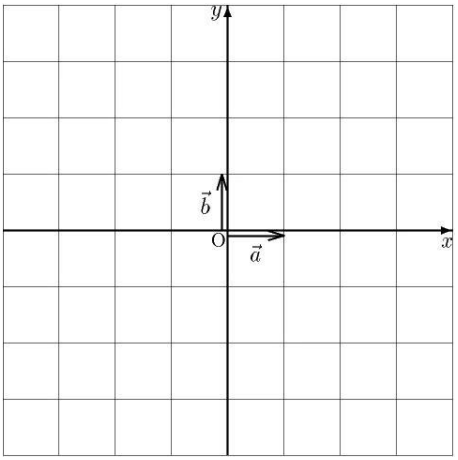


Linearly dependent



Linearly independent





BASIS VECTORS

A basis vector refers to a vector that forms part of a basis for a vector space. A basis is a set of linearly independent vectors that can be used to represent any vector within that vector space.

- **Vector Space (V):** Vector Space (V) is a mathematical structure of a set of vectors that can do addition and scalar multiplication. A set of vectors and operations that are defined on those vectors make up a mathematical structure called a vector space.

  ```
  Example: V = {(x, y) | x, y ∈ ℝ}
  ```

- **Field (F):** Field is the name of the scalar field over which the vector space V is defined. It offers the coefficients used in linear vector combinations. Common examples of fields are real numbers (R), complex numbers (C), and rational numbers (Q).
- **Basis (B):** A collection of linearly independent vectors that span the entire vector space V is referred to as a basis for vector space V.

Basis Vector

Basis vectors are the minimum set of linearly independent vectors needed to represent every vector in a vector space, without having unnecessary/ redundant vectors.

Every vector in the vector space can be expressed uniquely as a linear combination of the basis vectors.

There are **two important conditions** to be satisfied:

1. The vectors must be **linearly independent**.
2. They must **span the entire vector space**.

$$\text{Basis} = \text{Linear Independence} + \text{Linear Span}$$

Example: Basis of $\mathbb{R}^2$

Consider: $e_1 = (1, 0), \quad e_2 = (0, 1)$

These are called the **standard basis vectors**. Any vector (x, y) can be written as:

$(x, y) = x(1, 0) + y(0, 1)$

Therefore, they **span** $\mathbb{R}^2$. They are also linearly independent.

Basis vectors are building blocks of a vector space similar to LEGO blocks to construct the right different objects.

```
Basis vectors
      ↓
Basic building directions
      ↓
Linear combinations
      ↓
Different vectors
      ↓
Represent the entire space
```

Basis and Rank — Simple Explanation

These two concepts are closely connected:

Basis tells us the minimum building blocks needed to represent a vector space.
Rank tells us how many independent building blocks are actually present.

Rank: Rank is the number of linearly independent vectors in a matrix.

For a matrix, we can find rank by counting the number of **independent rows or independent columns**. The row rank always equals the column rank.

Example: Consider: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

The second row is just **2 × the first row**:

$$[2, 4] = 2[1, 2]$$

So the second row does not provide any new information. Therefore, only **one independent row** exists. So Rank (A) =1

For $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, Both rows are independent. So Rank (B) = 2

Basis	Rank
A set of vectors	A number
Provides building blocks	Counts independent building blocks
Must span the space	Measures dimension of the span
Must be linearly independent	Based on linear independence
Example: $\{[1, 0], [0, 1]\}$	Example: rank = 2

The number of vectors in any basis for that space equals the rank.

So Rank and Basis are important in AI for Dimensionality reduction, Removing Redundant features, PCA, and in Neural networks.

Linear Mapping

A **linear mapping** (or **linear transformation**) is a function that takes a vector as input and transforms it into another vector **while preserving the basic structure of vector operations**, making it fundamental to neural networks, PCA, Regression, Computer Vision and NLP.

Think of it as a **machine**:

Input vector → Linear Mapping → Output vector

For example: $\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} 2x \\ 3y \end{bmatrix}$

The mapping stretches the vector: twice in the x-direction and three times in the y-direction.

A mapping $T: V \rightarrow W$ is called a **linear mapping** if it satisfies two properties.

Property 1: Addition is preserved

$$T(u + v) = T(u) + T(v)$$

In simple words: Transform first and then add = Add first and then transform.

Property 2: Scalar multiplication is preserved

$$T(cu) = cT(u)$$

In simple words: If we multiply a vector by a number, its transformed version is also multiplied by the same number.

EX: Consider: $T(x, y) = (2, x3y)$
Take: $u = (1, 2),\ v = (3, 1)$
First add them: $u + v = (4, 3)$
Apply T: $T(4, 3) = (8, 9)$
Now transform separately: $T(u) = T(1, 2) = (2, 6)$ and $T(v) = T(3, 1) = (6, 3)$
Add: $T(u) + T(v) = (2, 6) + (6, 3) = (8, 9)$
Therefore: $T(u + v) = T(u) + T(v)$
So addition is preserved.

A linear mapping can: **Stretch** a vector; **Compress** a vector; **Rotate** a vector; **Reflect** a vector; **Shear** a vector.

But there is one very important restriction: **The origin must remain fixed.**

For a linear transformation: $\boxed{T(0) = 0}$
So if the origin is moved somewhere else, the transformation is **not purely linear**.

MOST IMPORTANT USES OF LINEAR MAPPING IN AI

AI Application	Why Linear Mapping is Used	Simple Example	Key Formula
Neural Networks	Converts input features into a new representation using weights	Student data → predicted performance	$z = Wx + b$
PCA / Dimensionality Reduction	Projects high-dimensional data onto fewer important directions	100 features → 10 principal components	$z = W^T x$
Linear Regression	Maps input features to a predicted numerical value	Area + bedrooms → house price	$y = Wx + b$
Computer Vision	Transforms image data/features into useful representations	Image pixels → feature representation	$y = Wx$
Embeddings / NLP	Transforms words or tokens into meaningful vector representations	Word → numerical vector	$y = Wx$

Affine Spaces:

A vector describes a direction or movement, while an **affine space describes points** that can be reached by starting from a particular point and moving in certain directions.

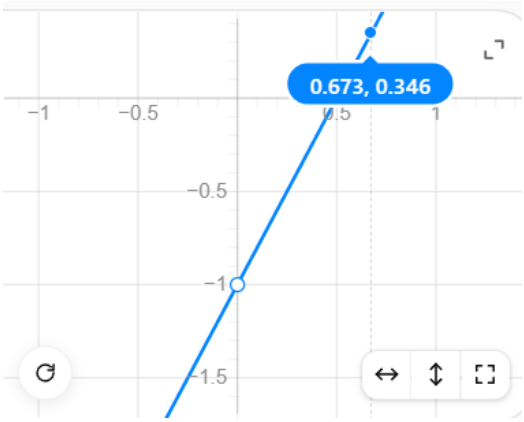
Start with a simple example

Suppose we have a point: $P = (2, 3)$ and a direction vector: $v = (1, 2)$
If we move along this direction, we get:

$P + tv$ where $t$can be any real number.

Therefore: $(x, y) = (2, 3) + t(1, 2)$
or $\boxed{(x, y) = (2 + t, 3 + 2t)}$. This represents a **straight line**.

Lets take y = 2x-1. On plotting we get the graph where line does not passes through the origin. Or the starting location does not have to be origin.



Vector Space vs Affine Space

Vector Space	Affine Space
Has a special origin	No special origin
Vectors can be added	Points are not normally added
Scalar multiplication is defined for vectors	We can move a point using a vector
Example: R^2	Example: a line not passing through origin
Describes directions and vectors	Describes positions and directions

An affine space can be written as:

$$\boxed{P = P_0 + v}$$

where:

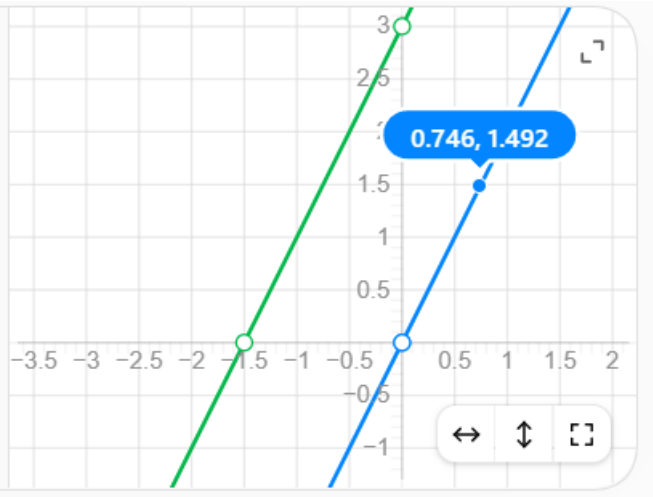
- P_0 = a starting/reference point
- v = a vector describing movement
- P = resulting point

For multiple directions:

$$\boxed{P = P_0 + c_1v_1 + c_2v_2 + \cdots + c_nv_n}$$

This is very similar to a linear combination, but we have a **starting point** P_0.

Linear subspace → must pass through the origin.
Affine space → does not have to pass through the origin.



Affine spaces are useful because **real-world data usually does not start at the origin**.